# class11

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## Background

We saw last day that the main repository for biomolecular structure (the PDB database) only has ~250,000 entries. UniProtKB (the main protein sequence database) has over 200 million entries!

## AlphaFold

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

##The EBI AlphaFold Database

The EBI AlphaFold database contains lots of computed structure models. It is increasingly likely that the structure you are interested in is already in this database. < https://alphafold.ebi.ac.uk/ > There are 3 major outputs for Alphafold 1. A model of structure in **PDB** format. 2. A **pLDDT score**: that tells us how confident the model is for a given residue in your protein (high values are good, above 70) 3. A **PAE score**: that tells us about protein packing quality.

If you can't find a matching entry for the sequence you are interested in AFDB you can run Alphafold yourself…

##Running AlphaFold We will use ColabFold to run Alphafold on our sequence https://github.com/sokrypton/ColabFold https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb

Figure from AlphaFold here!

```r
library(png)
library(grid)

img <- readPNG("HIV_MONOMER_94B5B_UNRELAXED_RANK_001_ALPHAFOLD2_PTM_MODEL_4_SEED_000.PDB-HIV_

grid.newpage()
grid.raster(img)
```

### Interpreting results

Custom analysis of resulting models We can read all the AlphaFold results into R and do more quantitative analysis than just viewing the structures in Mol-star:

Read all PDB models:

```r
library(bio3d)
```

```r
results_dir <- "hivprdimer_23119/"

Pdb_file <-  list.files(path = results_dir,pattern = ".pdb", full.names = TRUE)
print(Pdb_file)
```

```
[1] "hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.p
[2] "hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.p
[3] "hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.p
[4] "hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.p
[5] "hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.p
```

```r
pdbs <- pdbaln(Pdb_file, fit = TRUE, exefile="msa")
```

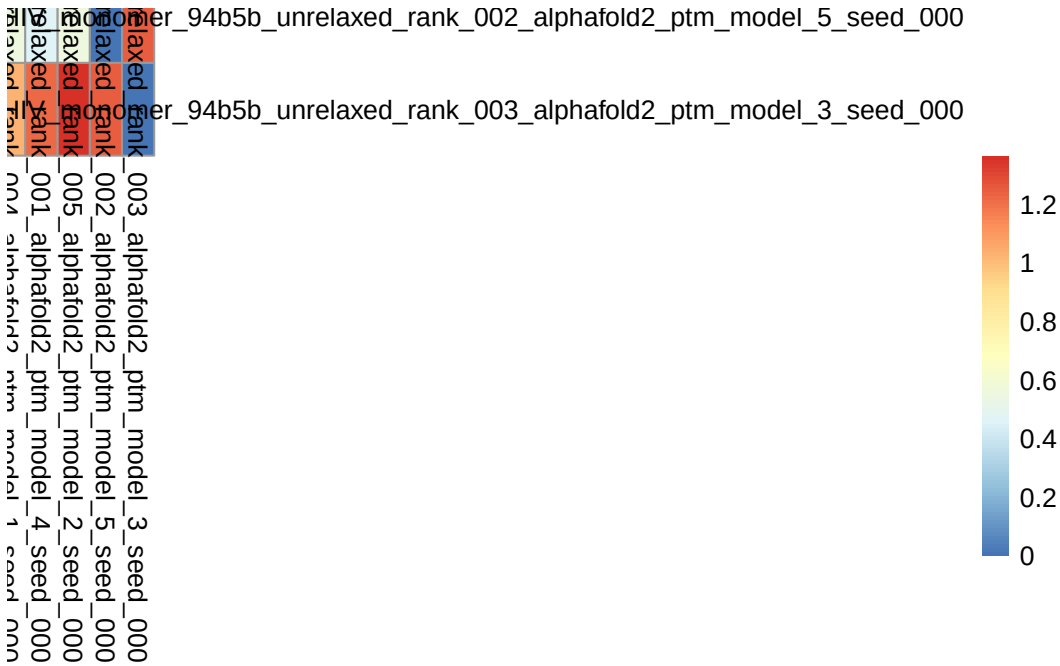
```
Reading PDB files:
hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb
hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb
hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb
hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.pdb
hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_001_alphafold2_ptm_mode
pdb/seq: 2   name: hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_002_alphafold2_ptm_mode
pdb/seq: 3   name: hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_003_alphafold2_ptm_mode
pdb/seq: 4   name: hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_004_alphafold2_ptm_mode
pdb/seq: 5   name: hivprdimer_23119//HIV_monomer_94b5b_unrelaxed_rank_005_alphafold2_ptm_mode
```

How similar or different are my models?

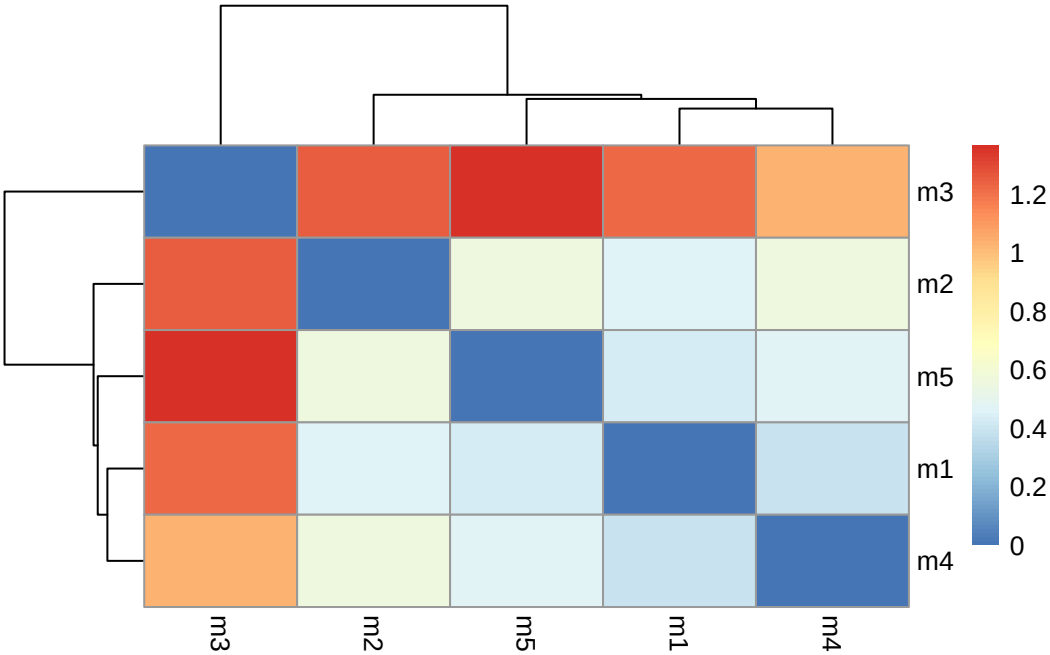
```r
rd <- rmsd(pdbs, fit = TRUE)
```

```
Warning in rmsd(pdbs, fit = TRUE): No indices provided, using the 99 non NA positions
```

```r
library(pheatmap)
pheatmap(rd)
```




```r
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



Models 1 & 2 are very similar, and models 4 and 5 are more similar to each other, while 3 is very different from both.

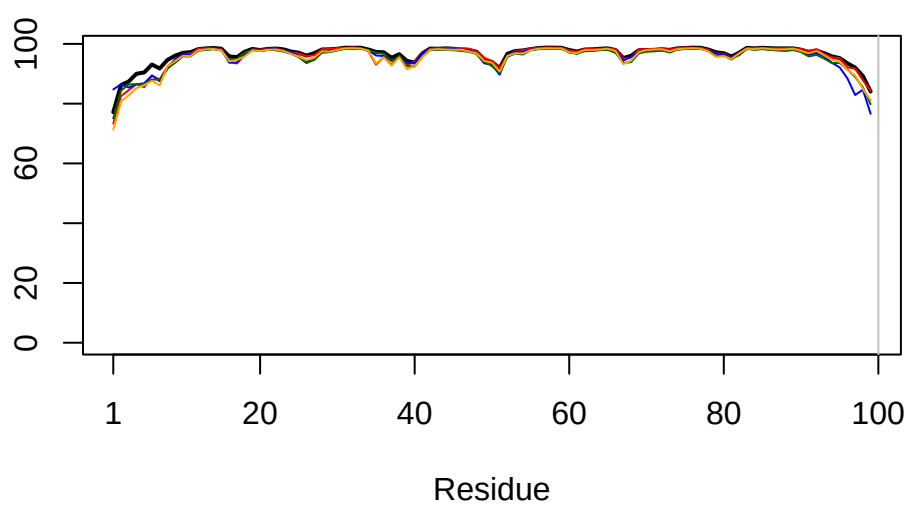
```
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
```

```
Warning in plotb3(pdbs$b[1, ], typ = "l", lwd = 2, sse = pdb): Length of input
'sse' does not equal the length of input 'x'; Ignoring 'sse'
```

```
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




pLDDT scores are generally high for most of the residues, meaning that there is a strong confidence that these parts are predicted correctly.

```r
core <- core.find(pdbs)
```

```
 core size 98 of 99  vol = 4.455
 core size 97 of 99  vol = 3.705
 core size 96 of 99  vol = 2.94
 core size 95 of 99  vol = 2.4
 core size 94 of 99  vol = 1.821
 core size 93 of 99  vol = 1.342
 core size 92 of 99  vol = 0.977
 core size 91 of 99  vol = 0.612
 core size 90 of 99  vol = 0.367
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

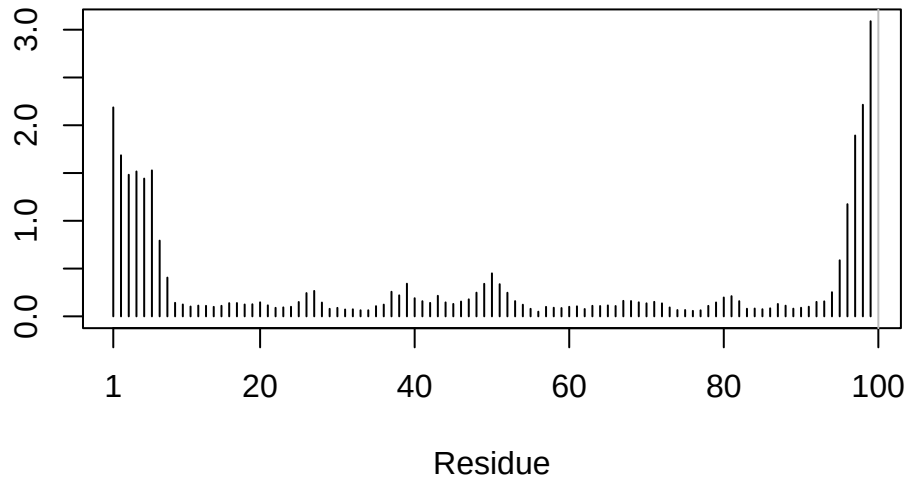
```
# 91 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     3   3      1
2     7  96     90
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
```

```
Warning in plotb3(rf, sse = pdb): Length of input 'sse' does not equal the
length of input 'x'; Ignoring 'sse'
```

```r
abline(v=100, col="gray")
```



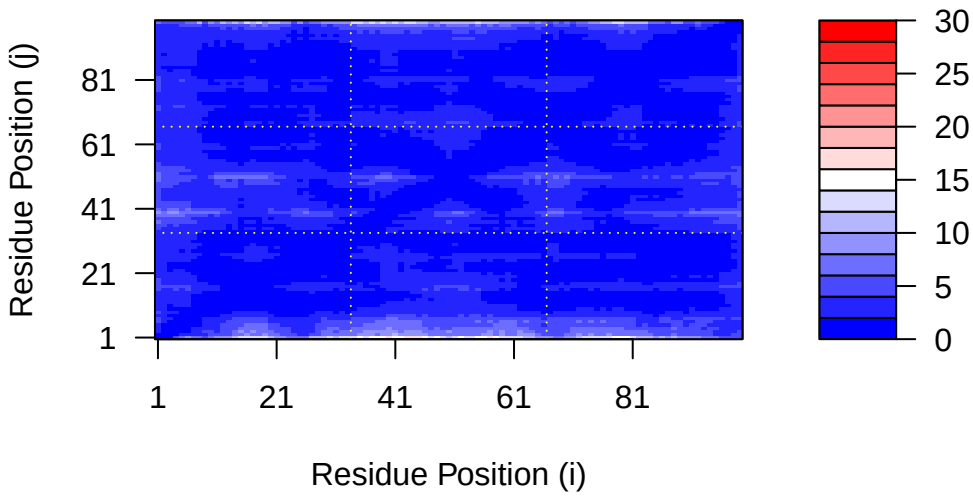

```
library(jsonlite)

pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)

pae1 <- read_json(pae_files[1], simplifyVector = TRUE)
pae5 <- read_json(pae_files[5], simplifyVector = TRUE)

pae1$max_pae
```

```
[1] 17.35938
```

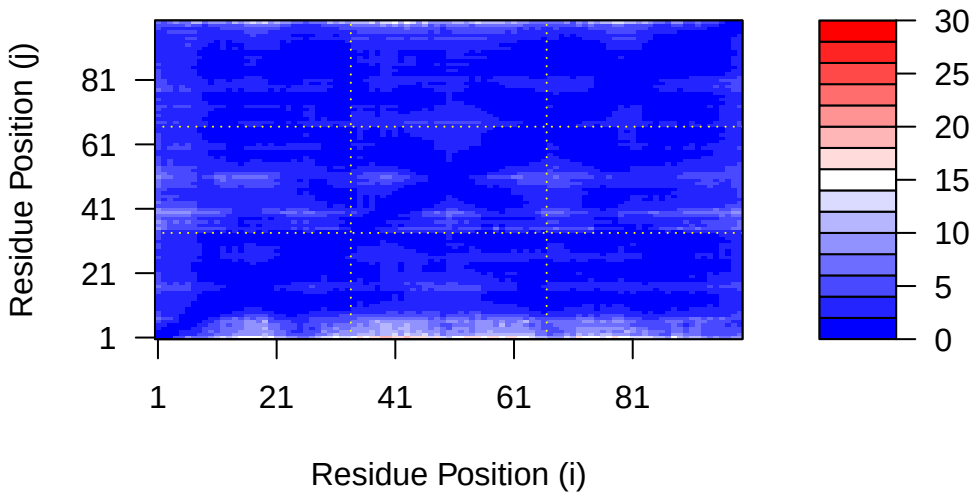
```
pae5$max_pae
```

```
[1] 19.51562
```

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          zlim=c(0,30))
```




```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          zlim=c(0,30))
```


Residue Position (j)
81
61
41
21
1
1
21
41
61
81
Residue Position (i)
30
25
20
15
10
5
0